

SQL Crash Course – Quick Reference (Kenya Health Informatics Context)

This 2-page cheat sheet covers the essentials you might need if a technical angle comes up in your interview. Focus on joins, grouping, and filtering.

1. Basic Structure

```
SELECT column1, column2
FROM table_name
WHERE condition
ORDER BY column1 DESC;
```

2. Joins

```
-- INNER JOIN: only matching rows
SELECT a.id, b.name
FROM patients a
INNER JOIN visits b ON a.id = b.patient_id;

-- LEFT JOIN: all rows from left + matches
SELECT p.id, v.date
FROM patients p
LEFT JOIN visits v ON p.id = v.patient_id;

-- RIGHT JOIN: opposite of LEFT JOIN
-- FULL OUTER JOIN: all rows from both tables
```

3. Aggregates

```
SELECT patient_id, COUNT(*) AS total_visits, AVG(age) AS avg_age
FROM visits
GROUP BY patient_id
HAVING COUNT(*) > 3;
```

4. Subqueries

```
SELECT name
FROM patients
WHERE id IN (
    SELECT patient_id
    FROM visits
    WHERE visit_date > '2025-01-01'
);
```

5. CRUD Basics

```
INSERT INTO patients (name, age) VALUES ('Jane Doe', 34);
UPDATE patients SET age = 35 WHERE id = 10;
DELETE FROM patients WHERE id = 10;
```

6. Constraints & Keys

- Primary Key – unique row identifier (e.g., id)
- Foreign Key – links between tables (e.g., patient_id)
- Unique, Not Null, Default – enforce data quality

7. Indexes

Indexes make lookups faster but slow down inserts/updates. Use on columns often in WHERE or JOIN.

8. NULLs

```
WHERE column IS NULL  
WHERE column IS NOT NULL
```

9. Views

```
CREATE VIEW patient_summary AS  
SELECT patient_id, COUNT(*) AS visits  
FROM visits  
GROUP BY patient_id;
```

10. HIV/EMR Example

```
SELECT p.id, p.name, l.result  
FROM patients p  
JOIN labs l ON p.id = l.patient_id  
WHERE l.test_type = 'Viral Load'  
      AND l.result > 1000  
      AND l.date > CURRENT_DATE - INTERVAL '6 months';
```

■ Master joins, grouping, filtering, and NULLs. That's usually enough to look solid.